Cloud Computing for ABC Enterprise

Key Contributions to Business Automation

CLOUD COMPUTING FUNDAMENTALS
ASSESSMENT 1

PROF DR. DIVYA LEEKHA

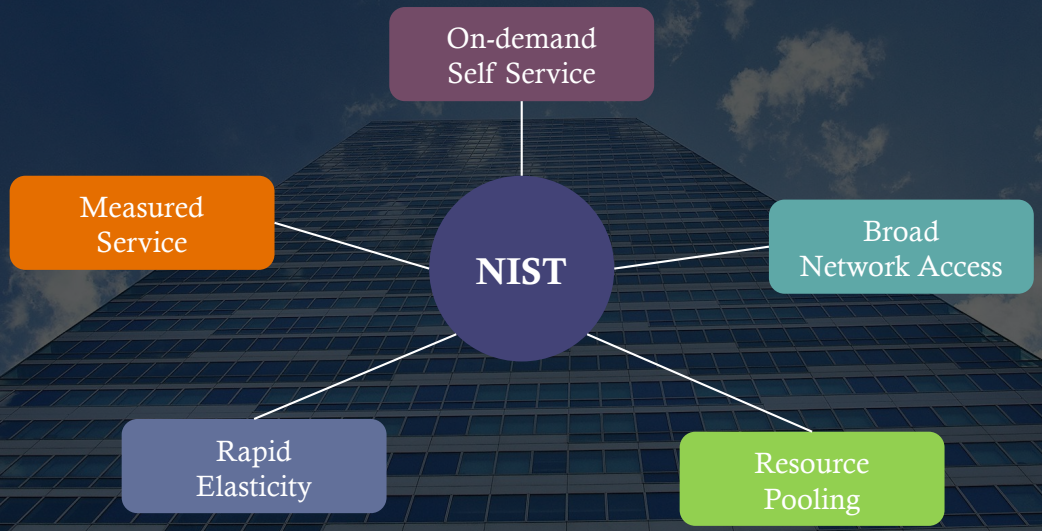
LUIS G. B. A. FARIA – ID A00187785

March, 2026

TORRENS UNIVERSITY AUSTRALIA

WHAT IS CLOUD COMPUTING?

On-demand delivery of compute, storage and services over the internet (Mell & Grance, 2011).



Key shift: from **owning capacity → renting capability**

TORRENS UNIVERSITY AUSTRALIA

BENEFITS OF CLOUD COMPUTING

B1 – Cost Efficiency

Traditional IT (CapEx)	Cloud – ABC (OpEx)
Large upfront hardware purchase	**No upfront servers**
Idle capacity during low demand	**Pay per compute hour.+ GB-month**
Full ops team required	**Provider manages infrastructure**
Fixed costs regardless of usage	**~80% reduction in start-up IT costs**

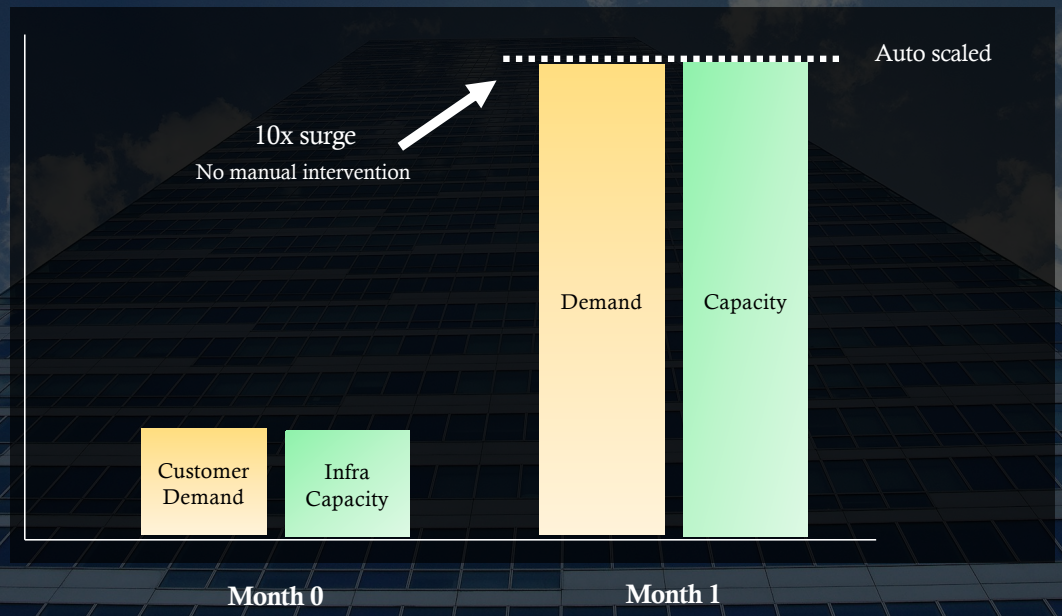
COST SHIFT

TORRENS UNIVERSITY AUSTRALIA

BENEFITS OF CLOUD COMPUTING

B2 & B3 – Scale without Limit & Less Overhead

Rapid Elasticity: ABC's 10x Growth absorbed by Auto Scaling



TORRENS UNIVERSITY AUSTRALIA

Challenges of Cloud Migration

Challenge	*Example*	*Mitigation*
Security & Privacy	PII + payment data	IAM least-privilege, MFA, encryption, shared responsibility model
Cost Volatility	Pay-as-you-go can spiral without guardrails	FinOps: budget alerts, resource tagging, rightsizing, reserved pricing
Vendor Lock-in + Skills Gap	Migration cost + mindset shift	Portability-first (containers, standard DBs), targeted cloud upskilling

TORRENS UNIVERSITY AUSTRALIA

Recommended Models for ABC

Cloud Service Models

IaaS
Compute + Config
Flexibility

PaaS
Managed DB, load balancing,
serverless

SaaS
Limited customization

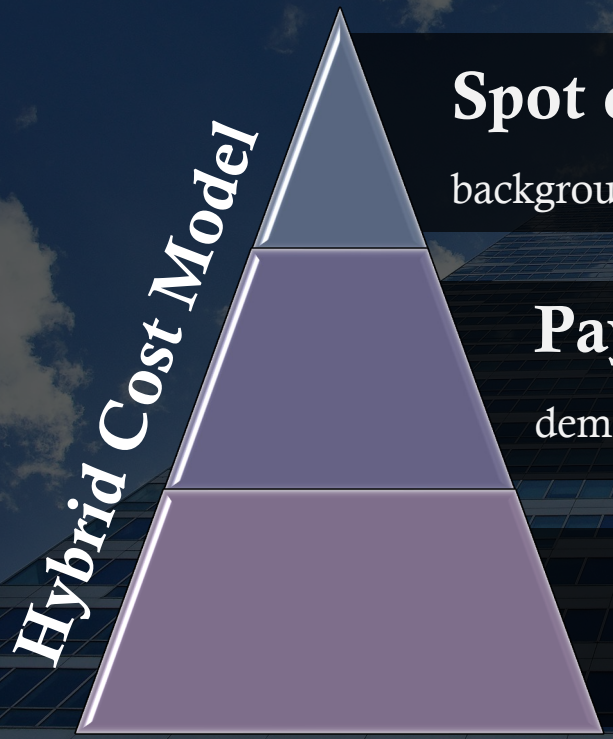
Cloud Deployment Models

Private cloud
High cost, limited elasticity

Hybrid cloud
Premature complexity

Public cloud
Scalable, automated,
cost-effective

Recommended Cost Model



Spot capacity

background jobs + analytics pipelines

Pay-as-you-go

demand spikes via Auto Scaling

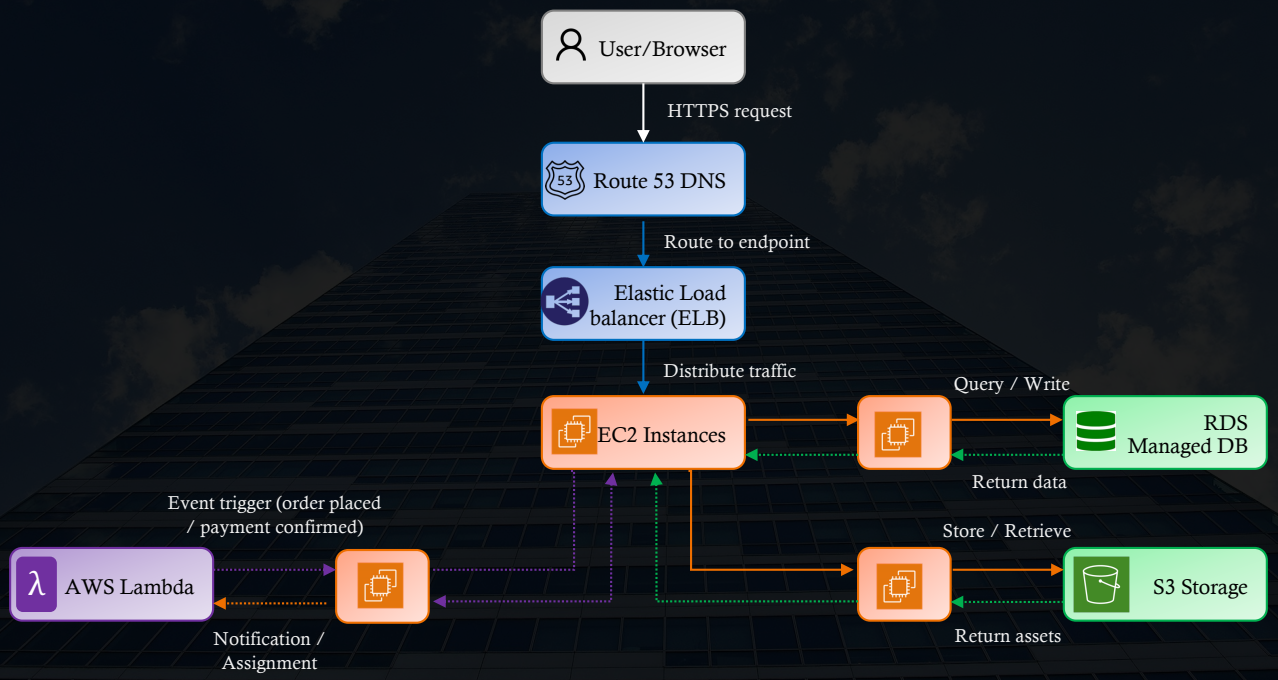
Reserved

stable tiers (web/app servers, databases)

WHY AWS?

Provider	Ecosystem Fit	Load Balancing	Serverless	ABC Alignment
aws	Broadest managed-service catalogue	ELB – native Route53 integration	Lambda – event-driven, zero idle cost	Best fit – Route 53 already in stack
Azure	Microsoft / enterprise-aligned	Application gateway – extra config needed	Azure Functions – strong but separate ecosystem	No existing Azure signals in ABC
Google Cloud	Analytics and ML-first	Cloud load balancing – GKE-oriented	Cloud Run / Functions – container first	No analytics-heavy workloads in ABC now

AWS IN ACTION



TORRENS UNIVERSITY AUSTRALIA

Conclusion

Summary of benefits for ABC

Next steps, final recommendations

TORRENS UNIVERSITY AUSTRALIA

REFERENCES

Alkhnbashi, O. S., Mohammad, R., & Hammoudeh, M. (2024). *Aspect-based sentiment analysis of patient feedback using large language models.* Big Data and Cognitive Computing, 8(12). https://doi.org/10.3390/bdcc8120167

Angelis, J. N., Murthy, R. S., Beaulieu, T., & Miller, J. C. (2024). *Better angry than afraid: the case of post data breach emotions on customer engagement.* IEEE Transactions on Engineering Management, 71, 2593–2605. https://doi.org/10.1109/TEM.2022.3189599

Bryman, A. (2016). *Social research methods* (5th ed.). Oxford University Press.

Cahya, V. N. C., Setyanto, R., & Paradise, P. (2025). *Analysis of the effectiveness of loyalty membership programs in increasing customer retention using net promoter score (NPS) with information system support*. Eduvest-Journal of Universal Studies, 5(9), 10974-10983.

Chen, E. (2023). *Growth product manager's handbook.* O'Reilly Media.

Cohen, J. (1988). *Statistical power analysis for the behavioral sciences* (2nd ed.). Routledge.

Creswell, J. W., & Plano Clark, V. L. (2023). *Designing and conducting mixed methods research* (4th ed.). SAGE Publications.

Dawes, J. G. (2024). *The net promoter score: What should managers know? International Journal of Market Research*, 66(2–3), 182–198. https://doi.org/10.1177/14707853231195003

Dawes, J. G. (2024). N*et promoter and revenue growth: An examination across three industries.* Australasian Marketing Journal, 32(1), 4-18.

Field, A. (2018). *Discovering statistics using IBM SPSS Statistics* (5th ed.). SAGE Publications.

Godovykh, M., & Pizam, A. (2023). *Measuring patient experience in healthcare.* International Journal of Hospitality Management, 112, 103405. https://doi.org/10.1016/j.ijhm.2022.103405

Mar, J., & Armaly, P. (2023). *Mastering customer success.* O'Reilly Media.

Morgan, D. L. (2014). *Pragmatism as a paradigm for social research*. Qualitative Inquiry, 20(8), 1045–1053. https://doi.org/10.1177/1077800413513733

Pallant, J. (2020). *SPSS survival manual* (7th ed.). Routledge.

Polonsky, M. J., & Waller, D. S. (2019). Quantitative data analysis. *In Designing and managing a research project* (4th ed., pp. 222-254). SAGE Publications. https://doi.org/10.4135/9781544316499

Shankar, R., & Yip, A. (2024). *Transforming patient feedback into actionable insights through natural language processing: A knowledge discovery and action research study.* JMIR Formative Research. https://doi.org/10.2196/69699

Wohlin, C., Runeson, P., Höst, M., Ohlsson, M. C., Regnell, B., & Wesslén, A. (2012). *Experimentation in software engineering*. Springer.

Wohlin, C., & Runeson, P. (2021). Guiding the selection of research methodology in industry–academia collaboration in software engineering. *Information and Software Technology*, 140, 106678. https://doi.org/10.1016/j.infsof.2021.106678

Xiao, Y., Li, C., Thürer, M., Liu, Y., & Qu, T. (2022). *Towards lean automation: Fine-grained sentiment analysis for customer value identification.* Computers and Industrial Engineering, 169. https://doi.org/10.1016/j.cie.2022.108186

TORRENS UNIVERSITY AUSTRALIA

Thank you!

CLOUD COMPUTING FUNDAMENTALS
ASSESSMENT 1

PROF DR. DIVYA LEEKHA

LUIS G. B. A. FARIA – ID A00187785

March, 2026

TORRENS UNIVERSITY AUSTRALIA